

Chapter – 1: Natural Resources and Their Use

Concerned social scientists are clear on what we need to do: we must move toward a regenerative economy, an economy that operates in harmony with nature, repurposing used resources, minimizing waste, and replenishing depleted resources. We must return to the innate wisdom of nature herself, the ultimate regenerator and recycler of all resources. — Christiana Figueres and Tom Rivett-Carnac in ‘The Future We Choose’

When does Nature become a Resource?

- Nature – meaning – total – life, non-life forms – part – env. – not created by humans
- Humans – use them – sustenance – create new things – consumption – these elements – resources
- Ex. – tree – part – env. – exist independently – humans – cut them – make furniture – view trees – resource
- These entities – not easily accessible
- Ex. – petroleum – present – places -deep under ocean – no tech. – access them – extraction cost – very high – culturally unacceptable – cutting trees – sacred groves
- Any entity – called resource – tech. accessible – exploitation – economic - culturally acceptable
- Usually – exploitation – negative connotation – this context – extraction, utilisation, consumption
- Earth – treasures – millions of yrs in making – humans – learnt to use
- These treasures – obvious – air, water, soil – not so obvious – coal, petroleum, precious stones, metal ores, timber
- Natural resources – materials, substances – occur – nature – valuable to humans

Categories of Natural Resources

- Science – learn – usefulness – categorization, naming – use – shared characteristics – categorize – ideas, things – name them – refer them – 1-word, short phrase
- These names, meanings – shared – grp. of people – helps – discuss – ideas, things – no need – describe them – every time
- Earlier classes – learnt – living, non-living things – as soon as – these words used – we know – their meaning
- One way – categorize natural resources – based on – their uses

Resources essential for life

- Life – no existence – without air, water, food
- Obt. these – atm., rivers, ponds, cultivation of soil, other living things
- Cannot make -air, soil, water

Resources for materials

- Humans – create – phy. obj. – nature’s gifts
- Make them – utility, beauty – enrich lives – ours, others
- Piece of wood – transformed – chair – carved into statuette
- India’s geographical diversity – provide us – variety – natural resources – wood to marble, coal to gold

Resources for energy

- Energy – cornerstone – modern living – electricity – buildings, transportation, production

- This energy – diverse natural src. – coal, water, petroleum, natural gas, sunlight, wind

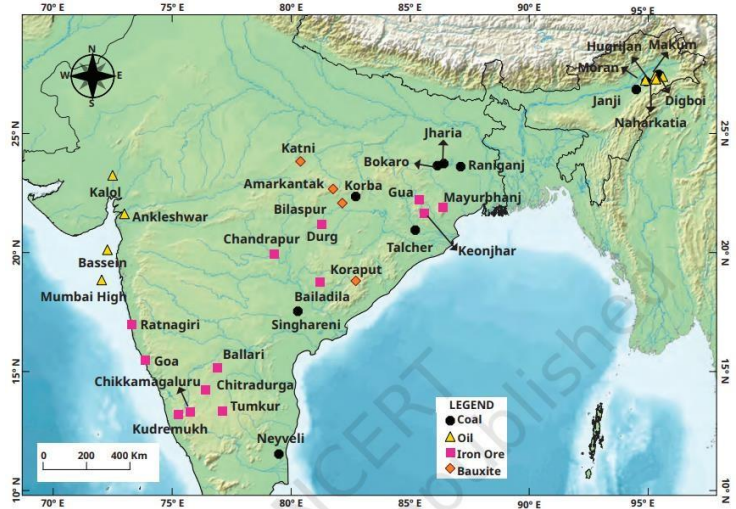
Renewable and non-renewable resources

- Second way – categorize natural resources – based on – renewable or not
- Gen. principle – nature – func. – restorative, regenerative way
- Restoration – process – returning something – original healthy state – if – damaged, degraded
- Nature – heals, renews, maintains – over time
- Cut on skin – heal normally – forest recover – wildfire
- Regeneration – beyond restoration – nature's ability – create new life – conditions – thriving
- Plant trees – areas – lost them – human interventions – clearing forest – housing
- Planting trees – originally grown there – restores ecosystem – trees – food, shelter – birds, squirrels, other creatures
- Nature – cycle – no waste
- Ex. – forest – tree falls – decomposes – bacteria, fungi, insects – part of soil – enrich it – new trees, plants – grow from seeds – some fall again – cycle restarts
- **Renewable resources**
 - Renewable resources – exhibit characteristics – over time
 - Most India – abundant sunshine – natural course – rivers fed – rain, melting ice – forest renew – soil replenish
 - Solar, wind, flowing water, timber – energy – renewable – if – managed – sustainable manner
 - To remain renewable – condition – natural rhythm – restoration, regeneration – not disturbed
 - Harvest timber – faster – forest regrowth – slower – forest deplete
 - Scientist – proved – irresponsible human actions – many cycles – disturbed
 - Combination – several factors – forest-fueled industrialization – cutting forests – agriculture, other purpose – increasing temp. – we experience
 - Result – glaciers – some places – Himalayas – melting – faster rate – precipitation – slower
 - Implications – water security – populations – plains – depend on – water tower
 - Need industries – produce goods – this process – creates waste – discharged – rivers, water bodies
 - Often – these wastes – cannot decay – become food – some lifeform
 - Instead – disturbance – nature's cycle – restoration, regeneration – situation – rivers – poisonous – no life
- **Non-renewable resources**
 - These resources – created – long periods – cannot replenish – rate we use
 - Ex. – fossil fuels – coal, petroleum – minerals – metals – iron, copper, gold – non-renewable
 - India – significant qty. – coal reserves
 - Mine coal – meet – growing needs – energy – BUT – coal reserves – 50 yrs. – cause – demands – increasing – population, development work – expand
 - Till – sustainable options – available widely – use current reserves – judiciously

Distribution of Natural Resources and its Implications

- Natural resources – no even distribution – across planet, country
- Uneven distribution – shape – human settlements, trade patterns, international relations, conflicts
- Many wars – fought – even continue – gain control – natural resources
- Industries – near natural resources – employment opportunities – local people
- Townships – grow – economic opportunity – expand

- More modern facilities – improve – quality of life – accessible
- BUT – these benefits – accompanied by costs – short term, long term
- Ex. – all over world – people – resource-rich areas – displaced from homes – facilitate developments
- Some cases – sacred places – under threat – conflicts
- National, international trade – depend on – geographical location – natural resources
- Combined – human skills, knowledge – unique products – Wootz steel
- Trade – fueled development – large empires – India
- Nature – no attention – political boundaries – leads – tensions – sharing – natural resources – states, countries
- Ex. – Kaveri river water – Karnataka, Tamil Nadu, Kerala, Puducherry
- Negotiations, management – req. – maintain peace, fair sharing
- Reaching – such agreements – neighbouring countries – not easy



The Natural Resource Curse

- Abundant natural resource – no guarantee – economic prosperity
- Some regions – rich – natural resources – experience – slower economic growth, development – economists call – **natural resource curse** – paradox of plenty
- Simple meaning – plenty – natural resources – does not mean – country – rich
- Many times – economies – unable – develop industries – convert resources – products – higher value – chapter – Factors of Production – latter part – more understanding
- India – avoided curse – invest – development – such industries – meet – growing needs
- BUT – challenge – balance resource extraction – sustainability – still remains
- Understanding, managing – natural resources – valuable start point – BUT – human knowledge, good governance, strategic planning – determine – lasting benefits – OR – temporary windfalls

Responsible and Wise Use of Natural Resources: Stewardship

- Sustain life – Earth – req. – respect nature – use resources – enable – restoration, regeneration – renewable resources – responsible, judicious use – non-renewable resources
- Scientists – study – these aspects – warned – irresponsible treatment – natural resources – pollution, **biodiversity loss**, climate change – taking place – increased pace – recent years

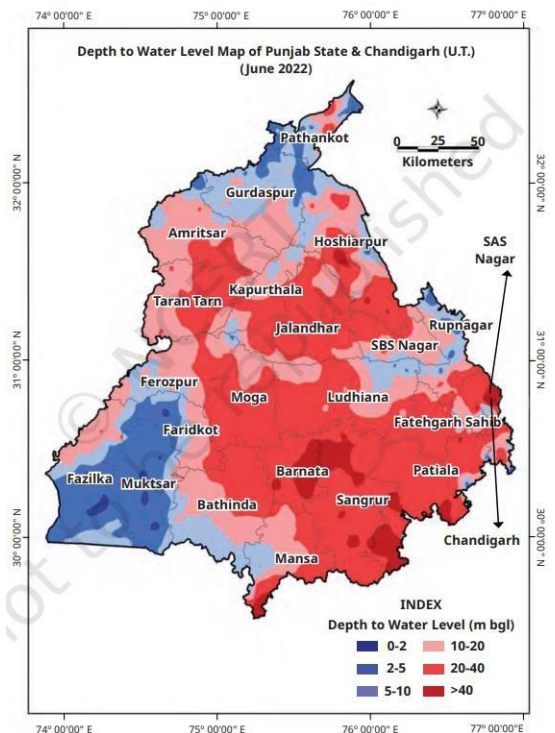
Restoration and regeneration of renewable resources

- Here – 2 examples – people – pushing – use – natural resources – beyond capacity – regenerate
- Many farmers – our country – use groundwater – irrigation
- Most states – extraction rate – faster than – replenishment rate – water tables
- This deficit – build up – higher cost – extraction – groundwater – eventually – unavailability
- Predicted – many cities – run out – groundwater – soon

- Initiatives – raise water tables – already taken
- Traditional methods – water harvesting – rejuvenation – ponds, tanks – cutting down – wasteful consumption – processing, reusing water – some strategies
- Similarly – improper use – chem. fertilisers, pesticides – soil degradation
- Traditional farming practices – consider soil – part – Mother Earth
- Practices – use – cow dung, natural fertilisers, mulching, multi-cropping – enable – soil mgmt.
- Must learn – these practices – apply them – current situation – prevent degradation – replenish, rejuvenate soil

- **Overexploitation of groundwater: a caselet from Punjab**

- Crisis – fertile plains – Punjab – groundwater – severely depleted
- Punjab – home – Green revolution – fed – large proportion – pollution – contribute – India – self-sufficiency – food
- Today – same state – issues – sustainability – Nature – exploited beyond regeneration – short term
- This concern – not particular – Punjab – affects – many other states
- 1960s – farmers – shift – high-yield varieties – wheat, paddy – req. – more water – extract groundwater – meet needs
- Addition – supply – free power – over-pumping – groundwater
- Modern farming tech. – req. – use – chem. pesticides, fertilisers
- Combined effect – these factors – groundwater – level – large parts – Punjab – inaccessible – depths – 30 m – ALSO – chem. – fertilisers, pesticides – dissolved – groundwater – health hazards
- Almost – 80% Punjab – classified – over-exploited
- Food security – ensured – short term – BUT – long term consequences – time, effort – heal



- **The case of cement**

- Cannot imagine life – without cement – houses, schools, hospitals, bridges, roads, airports – req. cement
- Production – cement – most polluting industries
- Production process – release – fine dust – enter lungs – damage them – settles – plant leaves – decrease yields – soil, water pollution
- Central Pollution Control Board – guidelines – cement factories – ensure – pollution – minimized, eliminated
- Additionally – shift – create alt. materials – reduce pollution
- Include – use – traditional materials – stone, mud, plant-based, recycled materials
- Traditional methods, materials – combined – modern tech. – create new materials – sustainable
- Production process – less pollution – local employment – designed – keep in mind – climate
- *Vrikshayurveda* – ancient Indian botanical science – focus – study, care – plants, trees
- Term – Sanskrit – *vriksha* – tree – *ayurveda* – science – life, health
- This sys. – date back – several millennia – formalized – texts – Surapala's *Vrikshayurveda* – 10th cent. CE

- Elaborate recommendations – specific plants – grown – diff. soil types – intricate methods – seed collection, preservation, pre-planting treatment
- Irrigation tech. – elaborate description – recommendations – vary – plant species, growth stage, seasonal conditions
- Also specify – pest mgmt. – natural repellents – plants – grow together
- This form – sustainable agriculture – promote – crop rotation, mixed cropping – maintain soil health
- Also offer – advice – proper methods – ploughing soil – retain moisture – facilitate growth – living org. – fungi, bacteria, earthworms
- **A caselet from Sikkim**
 - Pema's family farm – Sikkim – declining yields – mounting debts – expensive chem.
 - State govt. – policy – promote organic farming – their family – try it
 - Not easy transition – initially – yields drop – soil – recovering – yrs. – chem. use
 - Family – switch – compost – natural pest repellents – neem, garlic – multiple crops – across yr.
 - 5 yrs. later – Pema's farm – thriving – able – sell – cardamon, ginger, vegies – premium prices
 - 2016 – Sikkim – 100% organic state – all farmland – certified organic
 - Effects – transformational – local biodiversity – flourished – beneficial insects, birds – returned
 - Tourism – increased – visitors – obs. farming model – farmers' income – 20% +
 - Today – Sikkim – global model – demonstrate – entire region – successful transition – sustainable agriculture – improve – ecological, economic outcomes

Responsible and judicious use of resources

- Case – non-renewables – ensure – use resources – last long – humanity – find – sustainable alternatives
- Ex. – switch – renewable resources – energy – as many purpose as can
- **The International Solar Alliance – India's leadership in renewable energy**
 - India, France – International Alliance for Solar Energy (IASE) – 2015 – coalition – sunshine-rich countries – harness solar power
 - Alliance – focus – countries – abundant sunlight – yr. long
 - India – help – channel – billions of dollars – solar projects – developing nations – sharing – tech. expertise – create – affordable financing options
 - Bhadla solar park – symbol – India's solar ambitions – demonstrate – country – transition – traditional energy src. to renewable alternatives
 - Indians – this alliance – both – env. responsibility – economic opportunity
- We deal – these issues – be mindful – distribution, access – resources – often unfair – some sections – society
- Cities – many areas – not even – adequate, regular supply – drinking water
- Air pollution – industries, excessive use – fossil fuels – affect those – unable to protect – these hazards
- Must remember – rel. – nature, ourselves – act as steward – natural resources – restoration, regeneration, sustainability
- Bhagavad Gita – *lokasangraha* – idea – everyone – transcend personal desires – act – everyone's wellbeing